

Meme Stock Scanner

Testing Whether the Language Model and the Missing Data Are Needed

Quant research notes

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Abstract

This round tested two questions. First, whether the language model is needed at all, by running the pipeline with the language model removed from the decision path. Second, whether the data sources that are still missing are needed. The answer to the first is that the language model is not needed to reach a Sharpe ratio of 1.07 on the original universe. The answer to the second is that the missing data is needed, because the original universe list was produced by asking a language model to name potential meme stocks, and that list carried lookahead bias. When one of the missing data sources was added to build the universe from scratch, the system remained profitable at a Sharpe ratio of 0.61, but the alpha is not strong enough to deploy. The next requirement is social sentiment, and extracting it properly is where the language model belongs.

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1 Pipeline

The system has two layers with different update frequencies.

Layer 1, the universe. Runs once per day before the open. It answers which names are capable of a squeeze.

Layer 2, the ignition trigger. Runs on the candidate list only. It answers which of those names is starting to move now. Six conditions must all hold at the close of day t .

Position management. A hard stop, an early exit for positions making no progress, a volatility trailing stop, a trend break exit, and a maximum holding period.

Where the language model sits. The language model was placed in Layer 1 only, as an optional overlay that turns social text into a structured sentiment score. It was never given authority over entry or exit. For this round it was switched off entirely so the deterministic core could be measured on its own.

2 Parameter Choices and Their Justification

Every threshold below was either measured or derived. None was chosen by preference.

Parameter	Value	Why this value
Relative volume	2.0	A parameter sweep produced a Sharpe ratio between 0.83 and 0.91 across the range 1.75 to 3.0, and the ratio of the best cell to the median cell was 1.54, which indicates a plateau rather than a peak. The value 2.0 sits inside that plateau. Below 1.5 the signal is ordinary daily variation. Above 3.5 the Sharpe ratio falls to 0.40 to 0.60 because too few trades remain.
Exhaustion cap	50%	Measured by ablation. Total profit was \$434,670 with no cap, \$519,858 at 50%, and lower at 30% and 15%. Ignition days often gap more than 30% above the 20 day mean, and those are the largest winners, so a tighter cap removes them. A looser cap admits entries at the top of a completed move.
Momentum, days	5 5%	Rejects breakouts that are drifting sideways. Set low enough that it does not duplicate the relative volume condition.

Parameter	Value	Why this value
Hard stop	8%	These names have an average true range of 4% to 8% per day, so a stop tighter than 8% would be triggered by ordinary intraday movement. The realised average loss on stopped trades is 11.8%, not 8%, because positions gap through the level overnight.
Early exit	3% after 2 days	Measured by ablation. Exiting at 3% progress after 2 days added \$21,491 and reduced the count of winners above 50% only from 37 to 35. Exiting at 5% after 3 days lost \$41,692. Exiting at 10% after 3 days lost more. The 3% threshold is low enough to cut only positions that never moved.
Trailing stop	3 ATR	At 2 ATR the stop is inside the normal noise band of a stock moving 5% per day. At 3 ATR it survives the noise and still exits before a reversal completes.
Trend break	10 day mean	Applied only after a position is profitable. A break of the 10 day mean is the earliest reliable signal that an advance has ended.
Maximum hold	60 days	The original value was 10 days, which closed meme moves before they finished. Observed winners were held between 8 and 37 days, so 60 provides room without becoming an open ended position.
Position size	\$10,000	With a maximum of 10 concurrent positions this deploys the full \$100,000 of capital. Fixed size and no compounding mean the equity curve reflects the signal rather than the reinvestment schedule.
Liquidity floor	\$2M ADV	A \$10,000 order is 0.5% of a \$2M average daily volume, which is a participation rate that can be executed without moving the price. Below this level the impact cost model exceeds 1% one way.
Volatility floor	4% ATR	Measured. Without this floor the screen selects Canadian banks and utilities with dual listings, whose average true range is near 1.5%. Meme candidates sit between 6% and 10%.
Short interest lag	12 days	FINRA disseminates approximately 8 business days after the settlement date, which is about 12 calendar days. Using a shorter lag would consume information before it was public.
Universe size	150	Tested at 100, 150, and 200. The Sharpe ratio was 0.52, 0.61, and 0.53 respectively.

3 Pseudocode

```
# ----- LAYER 1, once per day, before the open -----
```

```

# Every input is as of the previous close. No value from day t is used.
for s in all_symbols:
    adv20 = mean(close[t-20:t] * volume[t-20:t])
    atr_pct = mean(true_range[t-20:t]) / close[t-1]
    dtc = finra_days_to_cover(s, asof = t - 12_days) # publication lag
    si_change = finra_short_interest_change(s, asof = t - 12_days)

    tradable = 2_000_000 <= adv20 <= 300_000_000 # 10k order is 0.5% of ADV
    volatile = atr_pct >= 0.04 # excludes banks and utilities
    has_short = dtc is not None

    if tradable and volatile and has_short:
        score[s] = 1.0*z(dtc) + 0.7*z(si_change) + 0.8*z(atr_pct)

universe = top_n(score, n = 150)

# ----- LAYER 2, at the close of day t -----
for s in universe:
    rvol = volume[t] / mean(volume[t-20:t])
    ma20 = mean(close[t-20:t])
    typ = (high[t] + low[t] + close[t]) / 3
    mom5 = close[t] / close[t-5] - 1
    ext = close[t] / ma20 - 1
    if (rvol >= 2.0 # inside the measured plateau of 1.75 to 3.0
        and close[t] > typ # close in the upper part of the range
        and close[t] > close[t-1] # rejects a one day pop
        and close[t] > ma20 # trend filter
        and mom5 > 0.05 # rejects sideways drift
        and ext <= 0.50): # 50% cap, measured optimum
        queue.append((rvol, s))

# ----- EXECUTION -----
for score, s in sorted(queue, reverse = True):
    if open_positions >= 10: break
    enter(s, price = open[t+1], size = 10_000) # never the close used to decide

# ----- POSITION MANAGEMENT, every session -----
for p in open_positions:
    stop = p.entry * 0.92 # 8%, wider than daily ATR of 4-8%
    if open[j] <= stop: exit(p, at = open[j]) # gapped through
    elif low[j] <= stop: exit(p, at = stop)
    elif age(p) >= 2 and mfe(p) < 0.03: exit(p, "no progress")
    elif close[j] <= peak(p) - 3 * atr14[j]: exit(p, "trail")
    elif close[j] > p.entry and close[j] < ma10: exit(p, "trend break")
    elif age(p) >= 60: exit(p, "max hold")

```

4 Hypothesis Under Test

Two questions were posed.

1. Is the language model necessary. A deterministic system is cheaper to backtest and easier to tune, so if the core performs without the language model, the language model can be deferred.
2. Is the missing data necessary. The system lacks real time borrow fee, utilisation, free float, dilution filings, and a historical social record.

The test runs in two stages. Stage one runs the pipeline with the language model removed on the original universe. Stage two widens the universe and asks whether the stage one result survives.

5 Stage One, Result Without the Language Model

Run with the language model switched off, on the original list of 82 names, from June 2020 to August 2026.

Metric	Value
Trades	1167
Win rate	36.1%
Expectancy per trade	+4.12%
CAGR	32.7%
Sharpe ratio	1.07
Sortino ratio	1.49
Maximum drawdown	−19.4%
Final equity	\$580,674

5.1 Why this result is good

A Sharpe ratio of 1.07 with a maximum drawdown of 19.4% is a respectable profile. The Sortino ratio of 1.49 is materially higher than the Sharpe ratio, which is expected for a strategy whose return distribution is skewed to the right. Sharpe penalises upside volatility, and this strategy produces most of its return through a small number of very large gains, so Sortino is the more informative of the two.

The breakdown by exit reason shows the exit ladder working as designed.

Exit reason	Trades	Win rate	Average return	Profit
Trend break, 10 day mean	323	100.0%	+30.4%	\$981,781
Volatility trailing stop	20	60.0%	+41.0%	\$81,975
Maximum hold	5	40.0%	+0.6%	\$275
No progress after 2 days	419	20.0%	−2.6%	−\$109,988
Hard stop at 8%	400	0.0%	−11.8%	−\$473,368

Three observations support the design. The trend break exit averages +30.4% and never closes a losing position, which means the rule that lets a trend run is doing exactly that. The no progress exit costs only 2.6% on average, so it clears dead positions at close to zero cost. The hard stop loses 11.8% on average against a nominal 8%, which is the honest cost of overnight gaps in this class of stock, and is the reason the stop is modelled on the low and the open rather than on the close.

The language model contributed nothing here because it was switched off. The first question is therefore answered. The language model is not required to reach this level of performance, and deferring it was the correct sequencing decision.

5.2 Why this result is not good

First, the profit is extremely concentrated. The 10 best trades account for 71% of total profit. The remaining 1157 trades account for 29%.

Second, the results are unstable across years.

Year	Trades	Win rate	Profit
2020	101	51.5%	\$114,657
2021	159	37.1%	\$98,956
2022	135	28.1%	−\$2,688
2023	229	34.5%	−\$1,839
2024	218	41.3%	\$281,869
2025	208	35.6%	\$33,329
2026	117	24.8%	−\$43,609

Two of 7 years are negative and 2024 alone produces 49% of the total. The strategy does not compound steadily.

Third, and most seriously, this run used the original list of 82 names, whose origin is examined in Section 6.

5.3 The 10 best trades

Ticker	Entry date	Entry	Exit date	Exit	Return	Days	Exit reason
LAES	2024-12-10	0.57	2025-01-07	7.15	+1152.2%	18	trend break
RGTI	2024-11-22	1.74	2025-01-08	10.04	+477.0%	30	trend break
COSM	2022-11-10	57.50	2022-12-06	235.45	+309.5%	17	trend break
PHUN	2021-10-20	52.50	2021-11-01	191.50	+264.8%	8	trend break
MARA	2020-07-24	1.04	2020-08-12	3.57	+243.3%	13	trend break
AMC	2021-05-13	127.70	2021-06-10	428.10	+235.2%	19	trend break
SOUN	2023-01-13	1.21	2023-02-16	3.66	+202.5%	23	trend break
ATER	2020-12-02	97.44	2021-01-27	277.44	+184.7%	37	trend break
LCID	2021-01-11	132.00	2021-02-23	352.10	+166.7%	29	trend break
ATER	2020-12-03	106.56	2021-01-27	277.44	+160.4%	36	trend break

All 10 exited on the trend break. None hit the trailing stop or the maximum hold. Holding periods run from 8 to 37 days. This confirms the exit ladder is correctly ordered, because the rule designed to let a trend run is the rule that closed every large winner.

5.4 The 10 worst trades

Ticker	Entry date	Entry	Exit date	Exit	Return	Days	Exit reason
COSM	2022-08-23	184.95	2022-09-20	112.60	−39.1%	19	hard stop
SNDL	2020-08-13	7.00	2020-08-14	4.49	−35.8%	1	hard stop
KSCP	2026-03-25	5.20	2026-03-30	3.39	−34.7%	3	hard stop
CENN	2026-04-07	7.43	2026-04-09	5.11	−31.2%	2	hard stop
HKD	2022-09-14	189.42	2022-09-15	134.69	−28.9%	1	hard stop
RBLX	2022-02-15	73.30	2022-02-16	53.87	−26.5%	1	hard stop
AMC	2026-06-18	2.83	2026-06-23	2.08	−26.5%	2	hard stop
ASTS	2023-06-21	6.29	2023-06-28	4.64	−26.2%	5	hard stop
BE	2025-01-24	29.81	2025-01-27	22.42	−24.8%	1	hard stop
KSCP	2024-08-14	14.45	2024-08-15	11.15	−22.8%	1	hard stop

All 10 exited on the hard stop and 7 of the 10 lasted 2 days or less. Realised losses range from 22.8% to 39.1% against a nominal stop of 8%. This is gap risk. The position opens below the stop

and fills at the open. No stop placement prevents this, so the only defence is position size, and this is the strongest argument in the study for keeping positions small.

5.5 Conclusion on the entry and exit logic

The entry and exit logic works as designed. The exit ladder is correctly ordered, the trend following exit captures the large moves, the no progress exit clears dead positions cheaply, and the hard stop limits damage subject to gap risk. The design is sound and does not require a language model. What it cannot do is choose which names to watch.

6 Where the Hindsight Bias Came From

6.1 The origin

At the start of the project a language model was asked to produce a list of potential meme stocks. It returned 82 names including GME, AMC, BBBY, LAES, GNS, and RGTI. That list became the universe, and the backtest then ran from 2020.

The problem is that a language model trained on data through 2026 already knows which stocks became meme stocks. Asking it for candidates does not produce a screen. It produces the answer. Every name on the list is there because of what it went on to do, so at every point in the simulation the system was watching a list that could only have been assembled after the period ended.

The three largest winners illustrate this directly. LAES returned +1152%, RGTI returned +477%, and COSM returned +309%. They are on the list precisely because they produced those moves.

This is not a coding error and no test conducted inside that universe can detect it. A null test against random entry timing was run and passed at $p = 0.025$, but that test held the universe fixed, so it only demonstrated that the timing rules beat random timing among names already known to have run.

6.2 How the bias was found

The bias was found by widening the universe. The same rules were applied to 113 randomly selected tradable names.

Universe	Trades	Win rate	Expectancy
List of 82 names from the language model	961	31.3%	+3.12%
Random 113 names, no screen	891	41.1%	−0.60%
Random plus volatility above 6%	202	27.7%	−3.00%
Random plus volatility and price cap	117	29.1%	−2.79%

Rules returning +3.12% per trade on the original list returned −0.60% on a neutral list. No structural filter rescued them.

The finding was confirmed at full scale on 10236 symbols with daily bars from 2016. Every admissible point in time definition returned a gross expectancy near +0.08% against a random control of +0.04%.

6.3 An isolated experiment

To confirm the difference came from the name list and nothing else, one variable was changed while entry, exit, sizing, costs, and data were held identical.

Method of choosing 82 names	Trades	Win rate	Expectancy	Sharpe	Final
List from the language model	1171	35.9%	+4.02%	1.06	\$570,696
Volatility based score	703	23.2%	−0.45%	0.45	\$68,111
Random selection	914	42.3%	+0.26%	0.26	\$123,401

The volatility based selector performed worse than random and overlapped with the original list in only 1 name. It was selecting delisting candidates rather than squeeze candidates. The problem was not that point in time selection is impossible. The selection features were wrong.

7 Removing the Bias by Adding the Missing Data

7.1 Screen version one, short interest

Short interest was one of the data sources previously listed as missing. Only a trailing 12 month window had been available. FINRA publishes the complete history, and the full market file was retrieved, giving 522,237 records across 5001 symbols and 131 settlement periods. Each observation is offset by 12 calendar days for the publication lag, so the screen is point in time. Names were ranked by days to cover and by the change in short interest.

Selection	Trades	Win rate	Expectancy	Sharpe	Final
List from the language model	1158	36.0%	+4.15%	1.08	\$580,529
Screen version one, top 100	1303	43.1%	+0.68%	0.52	\$189,034
Screen version one, top 150	1680	43.2%	+0.67%	0.61	\$212,817
Screen version one, top 200	1945	42.3%	+0.58%	0.53	\$212,487
Random 150	1594	43.6%	+0.15%	0.28	\$124,361

This is the first admissible construction in the study to clear its floor. Expectancy is +0.67% against +0.15% for random selection. The Sharpe ratio is 0.61 against 0.28. The maximum drawdown is 36.3% against 64.7%. Final equity is \$212,817 from \$100,000.

Adding one of the missing data sources therefore produced a result that is both profitable and free of lookahead bias. It also produced a much weaker one. A Sharpe ratio of 0.61 with a 36% drawdown and heavy profit concentration is not deployable.

7.2 Screen version two, correcting a measurement artifact

The names selected by version one are IMO, FTS, TRP, RY, and CM. These are Canadian banks, utilities, and pipelines with dual listings in the United States.

Their appearance is an artifact of how days to cover is computed. The measure is short interest divided by average daily volume, and for a dual listed security most trading occurs on the home exchange. The United States volume in the denominator is therefore small, which inflates the ratio. These names are not under squeeze pressure. They are thinly traded in one venue.

Version two removes the artifact by requiring an average true range of at least 4%. Canadian bank and utility shares move approximately 1.5% per day. Speculative small capitalisation names move 6% to 10%. The threshold separates the two groups.

Selection	Trades	Win rate	Expectancy	Final
Screen version one, top 150	1680	43.2%	+0.67%	\$212,817
Screen version two, top 100	894	28.7%	−2.41%	−\$115,082
Screen version two, top 150	1392	32.1%	−1.66%	−\$130,427
Screen version two, top 200	1729	34.8%	−0.75%	−\$30,205
Random within the same base	1773	40.1%	+0.02%	\$103,229

Version two now selects MVIS, SANA, HRTX, EOSE, LWLG, AVXL, RXRX, and OCGN. These are speculative biotechnology and technology small capitalisation names with high short interest and high volatility, which is the profile the strategy was designed around. Expectancy moved from +0.67% to −1.66%.

The interpretation is that the profitability of version one did not come from squeeze mechanics. It came from applying a momentum entry to liquid, slow moving, large capitalisation shares. When the screen is corrected to select the intended profile, the profit disappears.

Random selection within the same base, already filtered for liquidity, volatility, and the presence of short interest, returns +0.02%. The class as a whole shows no edge, and ranking within the class produces a worse result than not ranking at all.

7.3 A note on how these screens were evaluated

Coverage of the original 82 names was recorded during this work and is not reported as a success criterion here. Tuning a screen until it reproduces a list of known winners is circular, because that list was itself produced with knowledge of the outcomes. The volatility threshold in version two is justified only by the dual listing artifact described above. Any increase in overlap with the original list is a side effect of that correction and carries no evidential weight.

8 Limits of What Has Been Measured

Three limits apply to the signal measurements in this study and should be stated before any of them is used to justify further work.

The information coefficients were measured on a biased pool. Social and options signals were measured on the 195 name pool, of which 82 names came from the language model list. That pool is approximately 42% composed of names known to have squeezed, against a true base rate of roughly 1%. A coefficient measured there may not survive on a neutral universe. The measured values are reported below as observations, not as established properties.

Signal	IC at 20 days	Sample it was measured on
Drawdown from the all time high	+0.165	195 name pool, price data only
Prior meme history	−0.152	195 name pool, price data only
Out of the money call level	+0.061	110 names, 2024 onward
Social mention level	−0.054	159 names, 2021 to 2026
Social mention acceleration	+0.025	159 names, 2021 to 2026
Dollar volume surge, benchmark	+0.022	195 name pool
Short interest change, FINRA	near 0	195 name pool

Borrow fee and utilisation have not been measured at all. No historical source is available at any price the project has access to, so no coefficient exists for them. Statements about their importance rest on the mechanics of a short squeeze, not on evidence produced here. They are listed below as untested.

The social feature was deliberately crude. Only a count of ticker occurrences was stored. A count cannot separate 100 posts from 100 authors from 100 posts by 1 author, cannot separate a bullish thesis from a warning, and cannot identify whether a specific catalyst is named. Whatever the true value of social data is, the measurement so far understates or misstates it by an unknown amount.

9 Data To Be Tested Next

The correct conclusion from this study is not that a particular data source will supply the missing alpha. It is that the selection layer has been tested with only two of its possible inputs, price and short interest, and the remaining inputs have not been tested at all. The following list is ordered by expected information per unit of effort.

Data	What it would test	Availability
Social text, full universe	Whether attention acceleration selects names before a move, measured on a neutral universe rather than the enriched pool	Arctic Shift archive, free, rate limited, requires one overnight run per pass
Social depth, author identity and votes	Whether coordination can be separated from a single account posting repeatedly, which a mention count cannot do	Same source, requires storing raw text on a second pass
Options open interest and put volume	Whether gamma exposure can be estimated, and whether the call level coefficient of +0.061 survives outside the 110 name sample	Alpaca OPRA, from January 2024, already accessible
Free float and share count	Provides the denominator for short interest as a percentage of float, which is currently missing	SEC EDGAR quarterly filings, free, full history
Registered offerings and at the market programs	Whether dilution explains why some squeezes end early, which would be a negative filter	SEC EDGAR full text search, free, full history
Fails to deliver	An independent measure of settlement stress, already verified as retrievable but not yet integrated	SEC, free, from 2004, semi monthly
Borrow fee and utilisation	The only untested input with a direct mechanical link to squeeze pressure. No coefficient exists because no history exists	Ortex or Fintel, paid. A forward record started now is the only free path

Data	What it would test	Availability
Delisted securities	Whether results are inflated by the absence of names that went to zero	Polygon or Norgate, paid

10 Conclusions

1. The entry and exit logic is sound and does not need a language model. On the original universe it produced a Sharpe ratio of 1.07, a Sortino ratio of 1.49, a maximum drawdown of 19.4%, and final equity of \$580,674 from \$100,000.
2. The exit ladder is correctly ordered. All 10 of the best trades closed on the trend break exit at an average of +30.4%. All 10 of the worst closed on the hard stop. The no progress exit clears dead positions at an average cost of 2.6%.
3. Gap risk is the dominant execution risk. Realised losses on stopped trades average 11.8% against a nominal 8% stop, and the worst reached 39.1%. Small position size is the only defence.
4. That result is not durable. 10 trades produce 71% of the profit, 2 of 7 years are negative, and 2024 alone produces 49% of the total.
5. The bias entered through the universe. The list of 82 names was produced by asking a language model for potential meme stocks. A model trained through 2026 already knows which stocks became meme stocks, so the request returned the answer rather than a screen. No test conducted inside that universe can detect this, and one such test, a null test against random entry timing, passed at $p = 0.025$ while the universe was invalid.
6. Adding the missing short interest history removed the bias and retained a profit. Screen version one returns +0.67% per trade at a Sharpe ratio of 0.61 against +0.15% and 0.28 for random selection, with final equity of \$212,817.
7. That profit does not come from squeeze mechanics. Version one selects dual listed large capitalisation shares whose days to cover figure is inflated by thin United States volume. Correcting the artifact moves expectancy to -1.66% .
8. The selection layer has been tested with two inputs only, price and short interest. Social sentiment, options positioning, float, dilution, and borrow cost have either not been tested or have been tested only on a pool that carries the same bias the study set out to remove.
9. The strategy is therefore neither validated nor refuted. The execution layer is finished. The selection layer is untested against most of its candidate inputs.